Lesson 6.2 - Finding the Probability of a Binomial Distribution

> **Video:** Finding The Probability of a Binomial Distribution Plus Mean & Standard Deviation by The Organic Chemistry Tutor **Chapter:** 6 - Discrete Probability Distributions

Formula

$$P(X = x) = {}_nC_x \, p^x \, q^{n-x}, \qquad {}_nC_x = \frac{n!}{(n-x)!\,x!}$$

Symbol	Meaning
n	number of trials
x	number of successes you want (**exactly** x)
p	probability of success on one trial
$q = 1 - p$	probability of failure on one trial

Mean and standard deviation: $\mu = np, \quad \sigma = \sqrt{npq}$

Conditions: a fixed number of trials, each trial is success or failure, the trials are independent, and p is the same on every trial.

Example 1 - rolling a die

A die is rolled **12** times. What is the probability of getting a **4 exactly five times**?

- $n = 12$, $x = 5$, $p = \frac{1}{6}$ (one face out of six), $q = \frac{5}{6}$

Combination by hand:

$$_{12}C_5 = \frac{n!}{(n-x)!\,x!} = \frac{12!}{7!\,5!} = \frac{12 \cdot 11 \cdot 10 \cdot 9 \cdot 8}{5 \cdot 4 \cdot 3 \cdot 2 \cdot 1} = 11 \cdot 9 \cdot 8 = 792$$

Probability:

$$P(X = 5) = {}_nC_x \, p^x \, q^{n-x} = 792 \left(\tfrac{1}{6}\right)^5 \left(\tfrac{5}{6}\right)^7 \approx 0.0284 = 2.84\%$$

Example 2 - guessing on a multiple-choice test

A test has **20** questions, each with 4 choices (A-D) and one correct answer. A student guesses on every question. What is the probability of **exactly 6** correct?

- $n = 20$, $x = 6$, $p = \frac{1}{4} = 0.25$, $q = 0.75$

$$_{20}C_6 = \frac{n!}{(n-x)!\,x!} = \frac{20!}{14!\,6!} = \frac{20 \cdot 19 \cdot 18 \cdot 17 \cdot 16 \cdot 15}{6!} = 38{,}760$$

$$P(X = 6) = {}_nC_x \, p^x \, q^{n-x} = 38{,}760 \, (0.25)^6 (0.75)^{14} \approx 0.1686 = 16.86\%$$

(With 5 answer choices instead, $p = 0.20$ and $q = 0.80$.)

Example 3 - students taking algebra

25% of students at a school take algebra. **30** students are chosen at random.

- $n = 30$, $p = 0.25$, $q = 0.75$

(a) P(exactly 7 take algebra)

$$P(X = 7) = {}_nC_x \, p^x \, q^{n-x} = {}_{30}C_7 \, (0.25)^7 (0.75)^{23} = 2{,}035{,}800 \, (0.25)^7 (0.75)^{23} \approx 0.1662 = 16.6\%$$

(b) P(fewer than 5 take algebra), so $X < 5$, meaning $X = 0, 1, 2, 3, 4$ (5 is not included):

x	$P(X = x) = {}_{30}C_x (0.25)^x (0.75)^{30-x}$
0	0.000179
1	0.001786
2	0.008631
3	0.026853
4	0.060420
Sum	**0.09787 ≈ 9.79%**

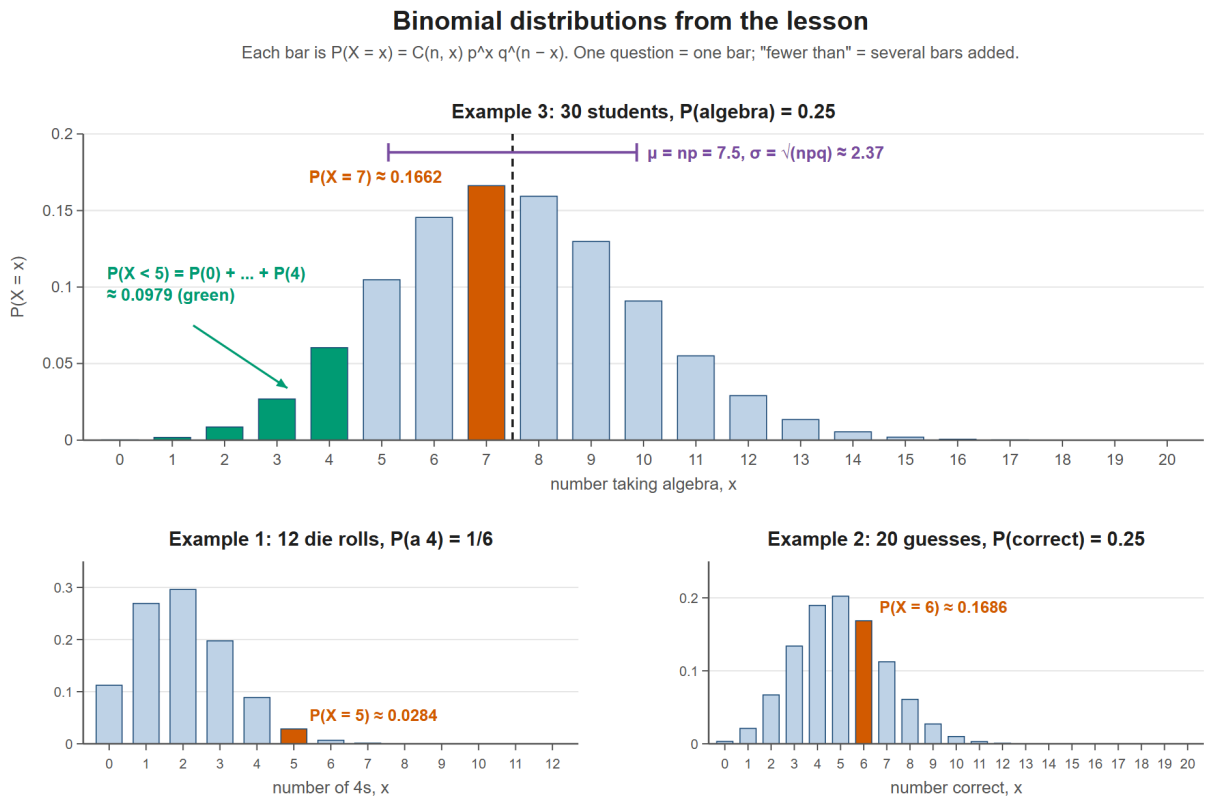
> **Correction:** the video adds only $x = 1$ to 4 and gets 9.77%. "Fewer than 5" also includes $x = 0$ (no algebra students at all), which adds 0.00018, so the correct total is about **9.79%**. The difference is tiny here, but leaving out $x = 0$ is a common mistake.

For large n, adding many terms gets tedious. A normal approximation, or the complement rule with a calculator's binomial CDF, is faster.

(c) Mean and standard deviation

$$\mu = np = 30(0.25) = 7.5$$

$$\sigma = \sqrt{npq} = \sqrt{30(0.25)(0.75)} = \sqrt{5.625} \approx 2.37$$



The full distribution behind each example. "Exactly 7" is a single bar. "Fewer than 5" is the sum of the five green bars, including the almost invisible bar at x = 0. The bars are tallest near the mean np = 7.5.

Key takeaways

- Identify n, x, p and q first, then substitute into ${}_nC_x p^x q^{n-x}$.
- "Exactly x" → one term. "Fewer than / at most / at least" → **add several terms**, and remember $x = 0$ when it applies.
- $\mu = np$ and $\sigma = \sqrt{npq}$.